Faculty led CAS

FISH2META RESULTS

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UWC ISAK Japan, class 2024
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Introduction

Our team

Fish2Meta is a team of 4 passionate students (17 years old) who aim to challenge themselves and dive deep in the world of scientific research.

- Lison (France): neuroscience
- Alessandro (Italy): coding
- Luka (Bosnia): maths
- Zelan (Philippines) : coding

Rationale

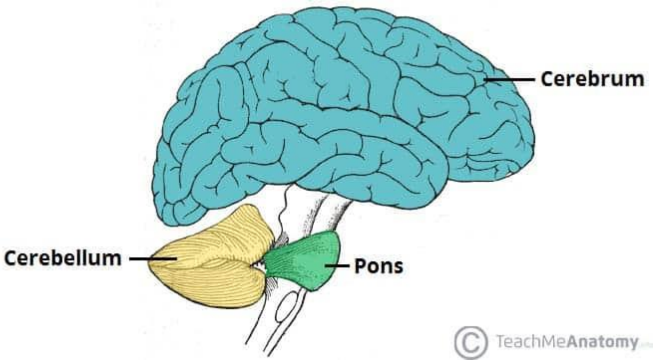


Fig1. *Image representing a cerebellum*

From preliminary research, we have learned that the cerebellum played an important role in the learning of individuals. This part of the brain is responsible for the acquisition and coordination of movements. It allows our bodies to perform incredibly complex movements involving several muscle pairs, timing, forces in almost an automatic manner such as throwing a ball to a target.

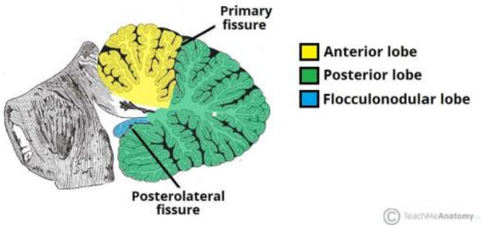


Fig2. *Image representing the different parts of the cerebellum*

Fig3. *Alessandro wearing prism adaptation googles*

Hence, we aimed to investigate whether the learning outcomes remain consistent when utilizing an adaptation prism. Additionally, we sought to explore the potential changes in adaptation and post effect patterns.

Martin et al. (1996)

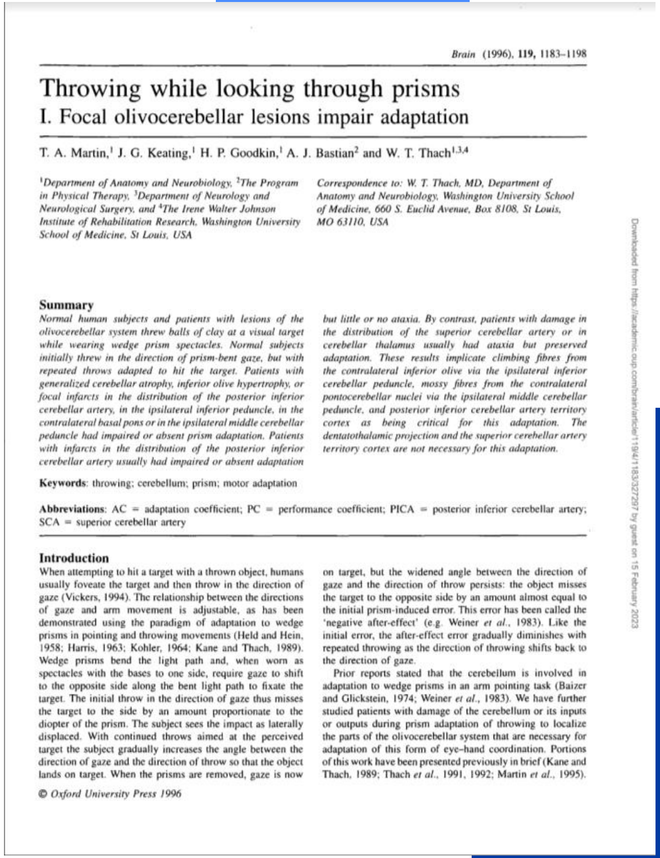
Inspiration
Prior reports stated that the cerebellum is involved in adaptation to wedge prisms in an arm pointing task. This paper studied patients with damage of the cerebellum or its inputs or outputs during prism adaptation of throwing to localize the parts of the olivocerebellar system that are necessary for adaptation of this form of eye-hand coordination.

Methods
Normal human subjects (control variable) and patients with lesions of the olivocerebellar system threw balls of clay at a visual target while wearing wedge 17º prism spectacles.

Evaluation Metrics

- Adaptation Coefficient (AC) - *rate at which adaptation takes place*
  AC is the rate of change of slope of the exponential decay curve of the subject throwing graphs.

- Performance Coefficient (PC) - *is the standard deviation of the horizontal error of the subject's last eight throws wearing prisms.*



Brain (1996), **119,** 1183–1198

Throwing while looking through prisms
I. Focal olivocerebellar lesions impair adaptation

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Summary
Normal human subjects and patients with lesions of the olivocerebellar system threw balls of clay at a visual target while wearing wedge prism spectacles. Normal subjects initially threw in the direction of prism-bent gaze, but with repeated throws adapted to hit the target. Patients with generalized cerebellar atrophy, inferior olive hypertrophy, or focal infarcts in the distribution of the posterior inferior cerebellar artery, in the ipsilateral inferior peduncle, in the contralateral basal pons or in the ipsilateral middle cerebellar peduncle had impaired or absent prism adaptation. Patients with infarcts in the distribution of the posterior inferior cerebellar artery usually had impaired or absent adaptation but little or no ataxia. By contrast, patients with damage in the distribution of the superior cerebellar artery or in cerebellar thalamus usually had ataxia but preserved adaptation. These results implicate climbing fibres from the contralateral inferior olive via the ipsilateral inferior cerebellar peduncle, mossy fibres from the contralateral pontocerebellar nuclei via the ipsilateral middle cerebellar peduncle, and posterior inferior cerebellar artery territory cortex as being critical for this adaptation. The dentatothalamic projection and the superior cerebellar artery territory cortex are not necessary for this adaptation.

Keywords: throwing; cerebellum; prism; motor adaptation

Abbreviations: AC = adaptation coefficient; PC = performance coefficient; PICA = posterior inferior cerebellar artery; SCA = superior cerebellar artery

Introduction
When attempting to hit a target with a thrown object, humans usually foveate the target and then throw in the direction of gaze (Vickers, 1994). The relationship between the directions of gaze and arm movement is adjustable, as has been demonstrated using the paradigm of adaptation to wedge prisms in pointing and throwing movements (Held and Hein, 1958; Harris, 1963; Kohler, 1964; Kane and Thach, 1989). Wedge prisms bend the light path and, when worn as spectacles with the bases to one side, require gaze to shift to the opposite side along the bent light path to fixate the target. The initial throw in the direction of gaze thus misses the target to the side by an amount proportionate to the diopter of the prism. The subject sees the impact as laterally displaced. With continued throws aimed at the perceived target the subject gradually increases the angle between the direction of gaze and the direction of throw so that the object lands on target. When the prisms are removed, gaze is now on target, but the widened angle between the direction of gaze and the direction of throw persists: the object misses the target to the opposite side by an amount almost equal to the initial prism-induced error. This error has been called the 'negative after-effect' (e.g. Weiner *et al.*, 1983). Like the initial error, the after-effect error gradually diminishes with repeated throwing as the direction of throwing shifts back to the direction of gaze.

Prior reports stated that the cerebellum is involved in adaptation to wedge prisms in an arm pointing task (Baizer and Glickstein, 1974; Weiner *et al.*, 1983). We have further studied patients with damage of the cerebellum or its inputs or outputs during prism adaptation of throwing to localize the parts of the olivocerebellar system that are necessary for adaptation of this form of eye–hand coordination. Portions of this work have been presented previously in brief (Kane and Thach, 1989; Thach *et al.*, 1991, 1992; Martin *et al.*, 1995).

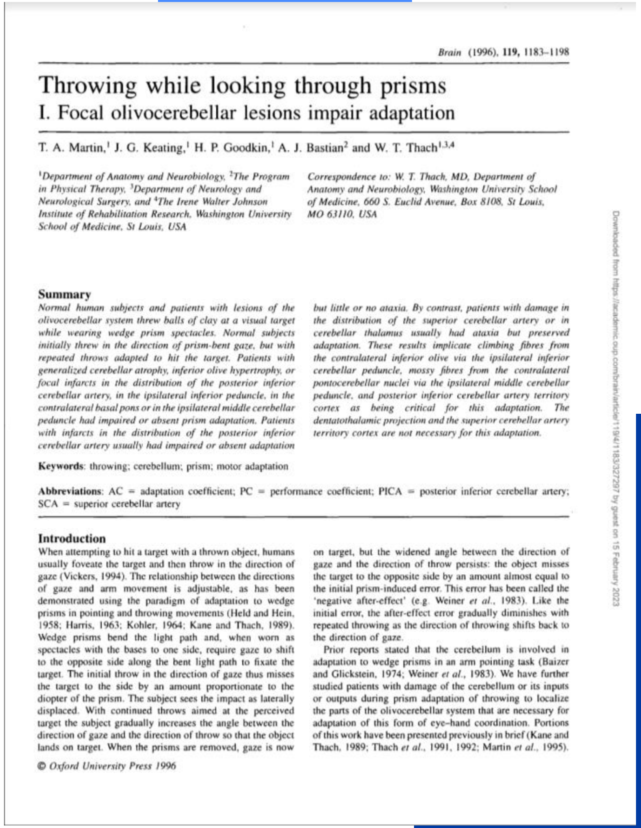
Martin et al. (1996)

Insights

- The paper sets a standard for a prism adaptation-related study including how data should be presented, contents of the paper, and experimental setup among others.
- The AC & PC values help us evaluate a subject's performance more precisely compared to simply graphing the points and evaluating the learning or the lack thereof visually.

PC $$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$ Standard deviation of last 8 throws during baseline

AC $$f(t) = a - be^{\frac{-t}{\mathbf{AC}}}$$ Rate of change of slope of the exponential decay curve of the subject throwing graphs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What do we want to know?

Q1

What changes occur when hand is changed after prism adaptation?

"The adaptation of right-handed throws does not transfer to left-handed throws; the adaptation of overhead throws usually does not transfer to underhand throws. Adaptation occurs gradually through practice. The process itself is not entirely conscious; it can occur in parallel with voluntary corrections."
- ***From the Dynamic Coordination of Body Parts During Prism Adaptation (Kohler 1964).***

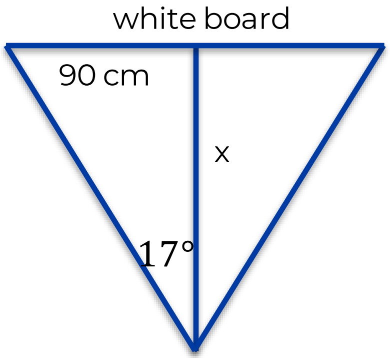
Are different ways of throwing the ball independent of each other when it comes to adaptation and the post effect?

Q2

Hypothesis

Different ways of throwing the ball are independent of each other. During the prism adaptation, one will not be affected by the other. Thus, participants will have to adapt to prism separately for each hand movement. Post effects will be independent of each other.

Experimental setup



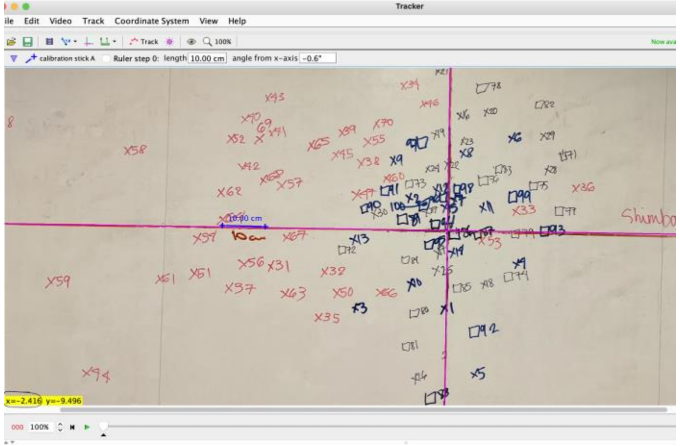
Calculation of the distance to the board

Participants stand 2.95 m from the board and throw a ball at the board. Their goal is to hit the centre.

The picture of the board is taken and the data is analysed using the Tracker application. Distance from vertical axis is measured.

Further analysis is done using Python.

Experimental setup



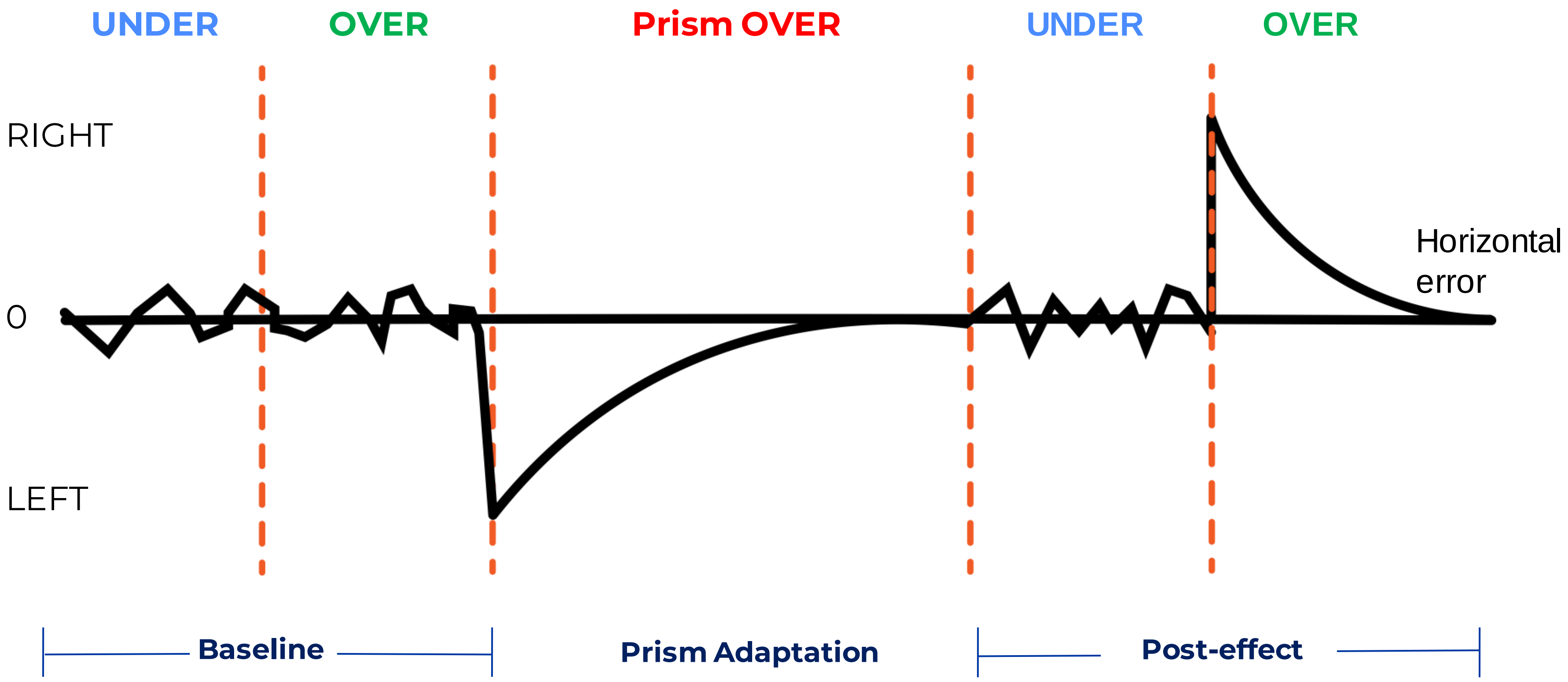
Tracker program

02
Our first Experiment

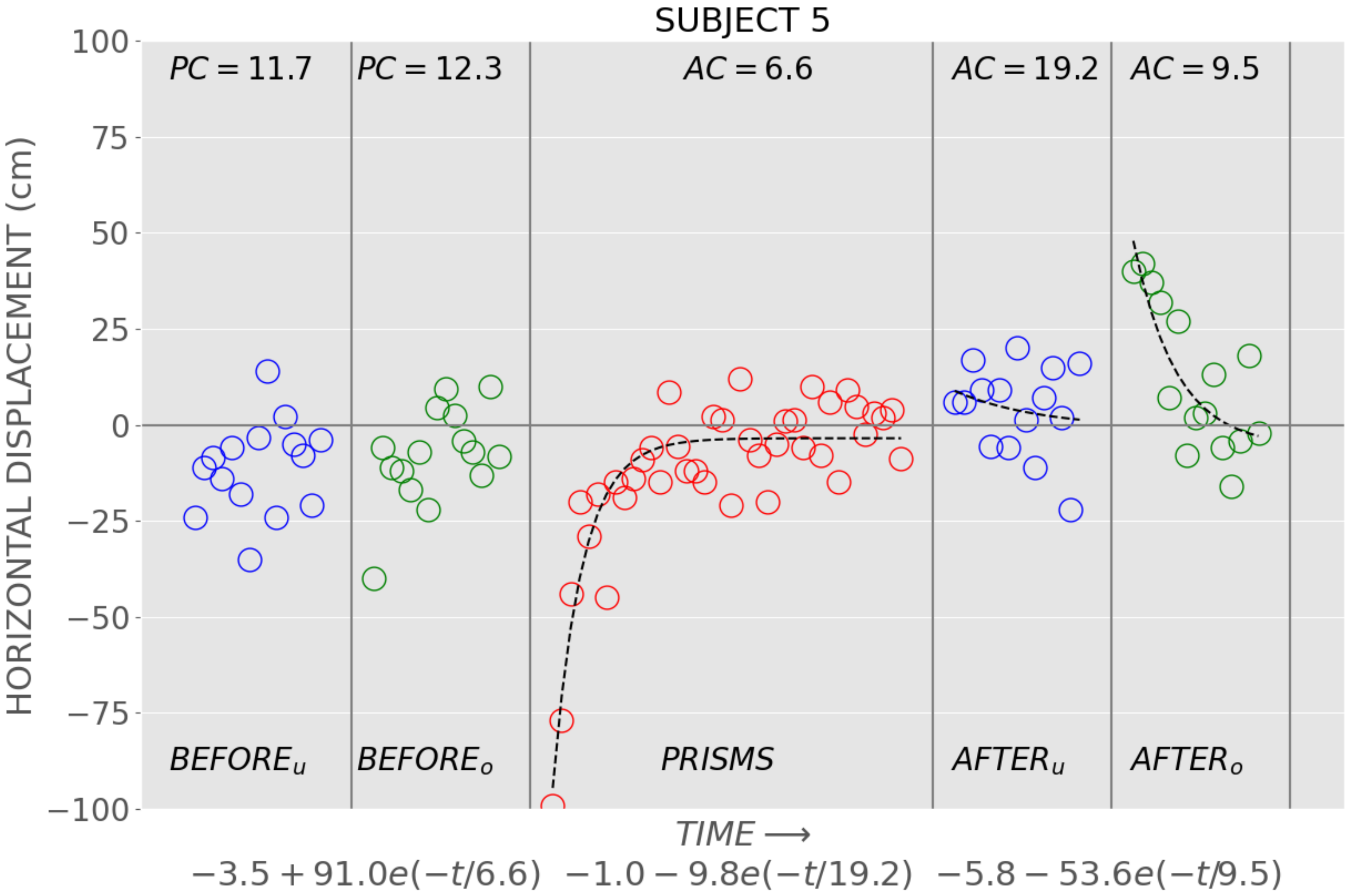
Experimental Protocol

Polyurethane foam ball ⌀ 9cm 4.5g

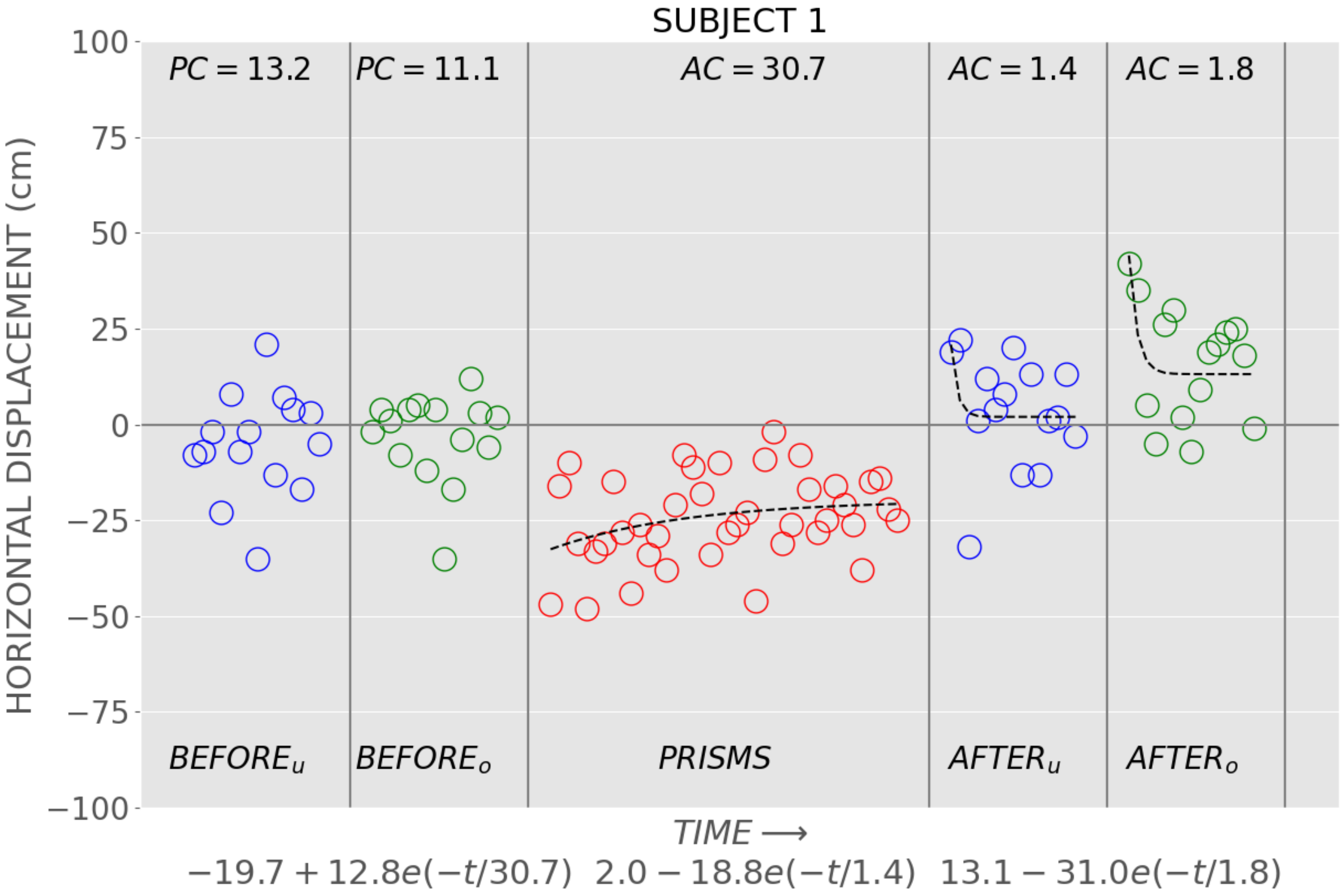
What do we predict we will see?



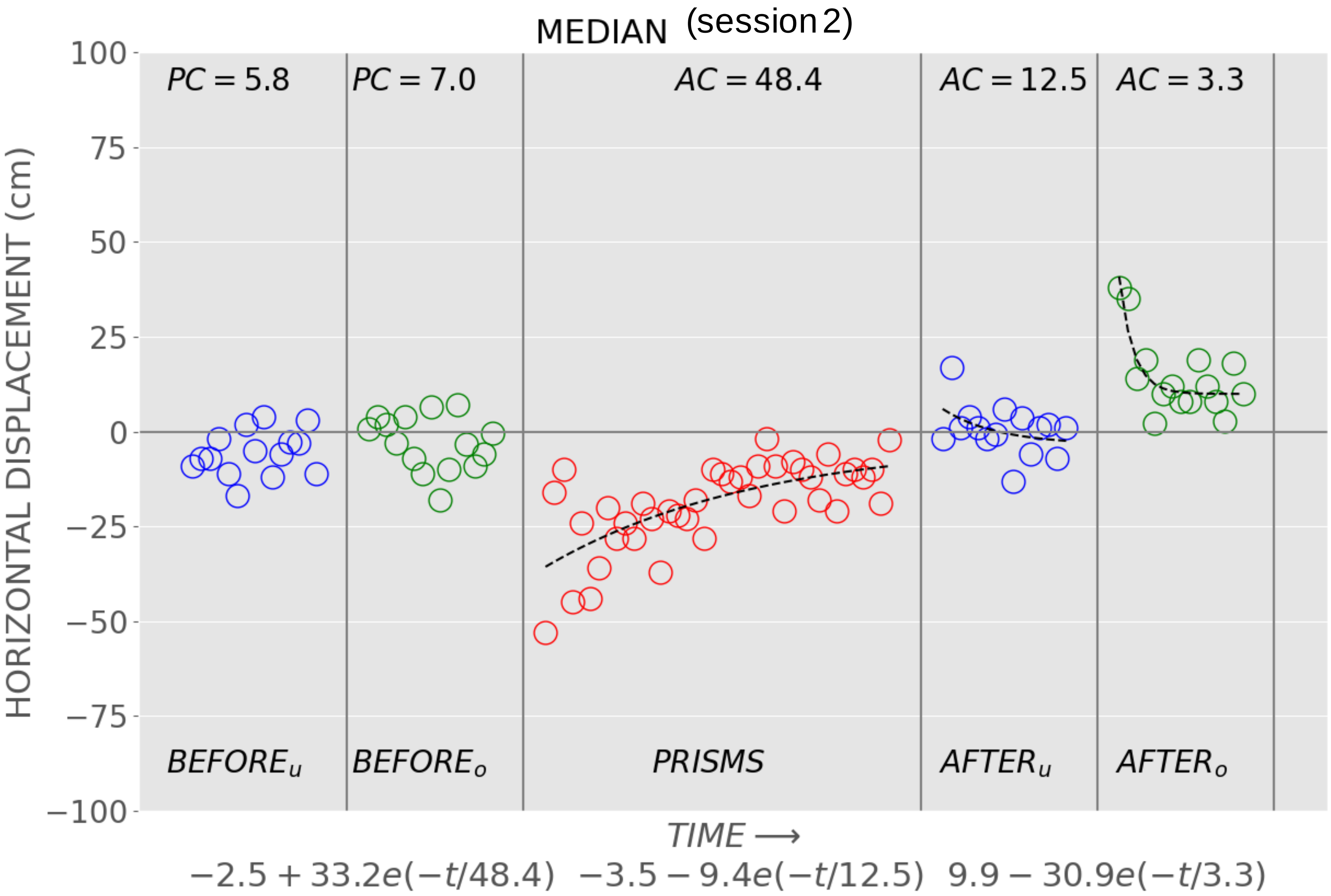
Results Session 2



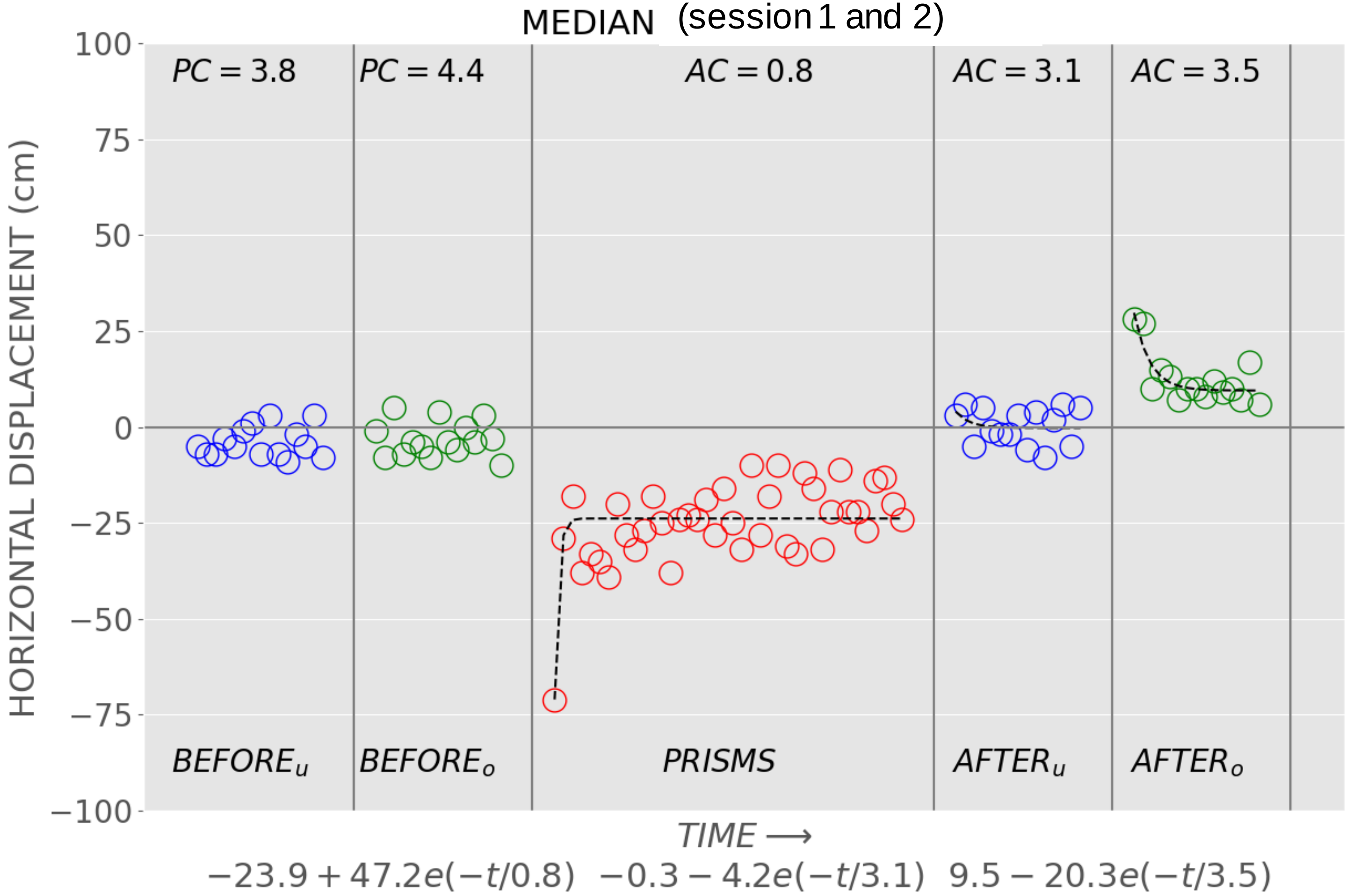
Results Session 2



Median results session 2 (n=4)



Median results both sessions n=9



Conclusion

- The experiment did not utilize an underhand throw during the prism adaptation phase. Thus, we can see that the rates of learning for both the baseline and post-effect phase where an underhand throw was utilized, are the same.

- The experiment used an overhead throw during the adaptation phase. In contrast to the post-effect with the underhand throw, there is visible readaptation.

- These results support our hypothesis: "Different ways of throwing the ball are independent of each other. During the prism adaptation, one will not be affected by the other. Thus, participants will have to adapt to prism separately for each hand movement."

03

Experiment 2

Experimental Protocol 2

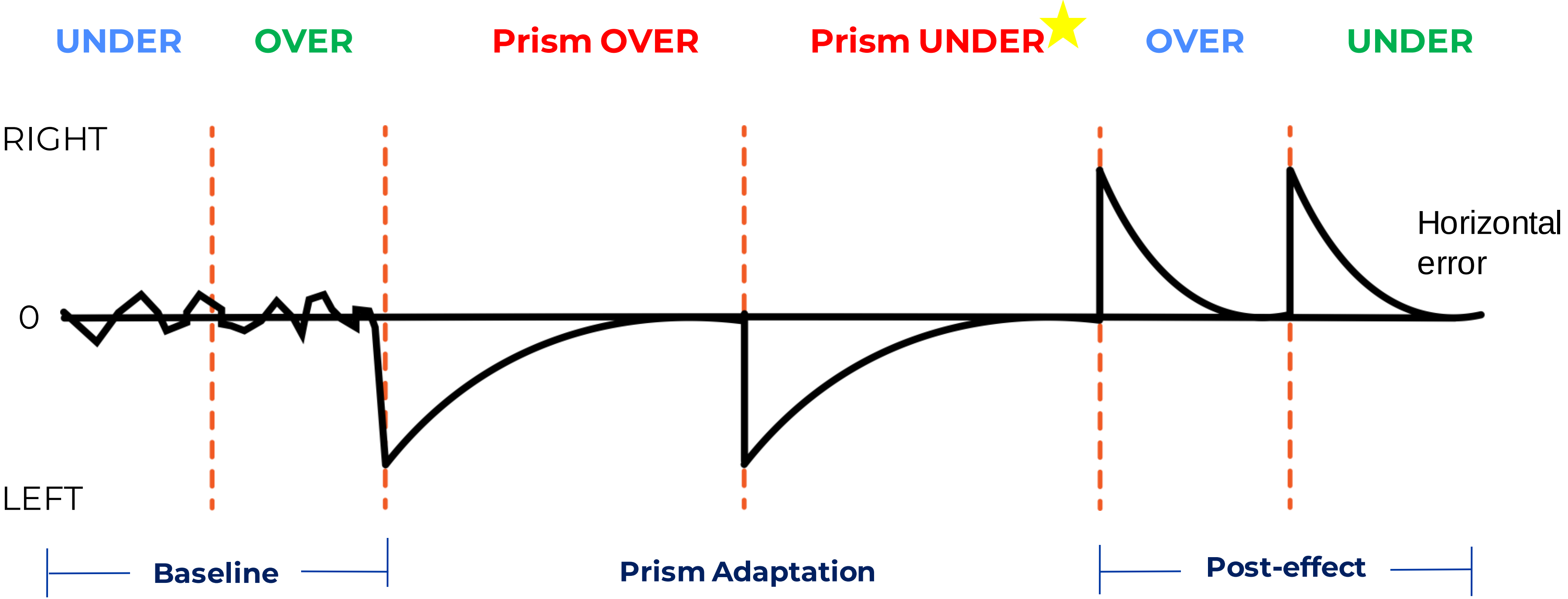
Polyurethane foam ball ⌀ 9cm 4.5g

Throws:

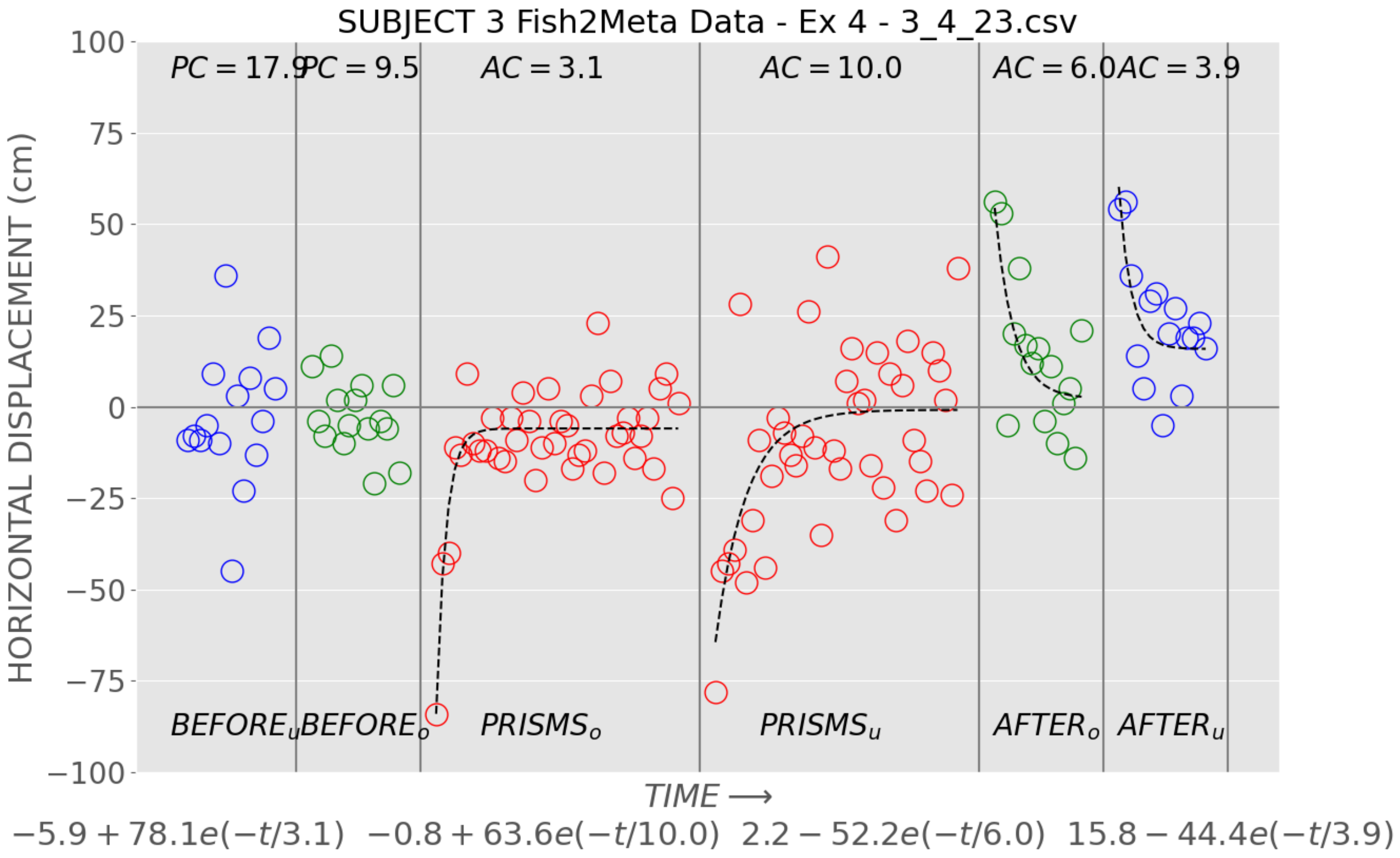
x15 x15 x40 x40 x15 x15

Baseline ***Prism Adaptation*** ***Post-effect***

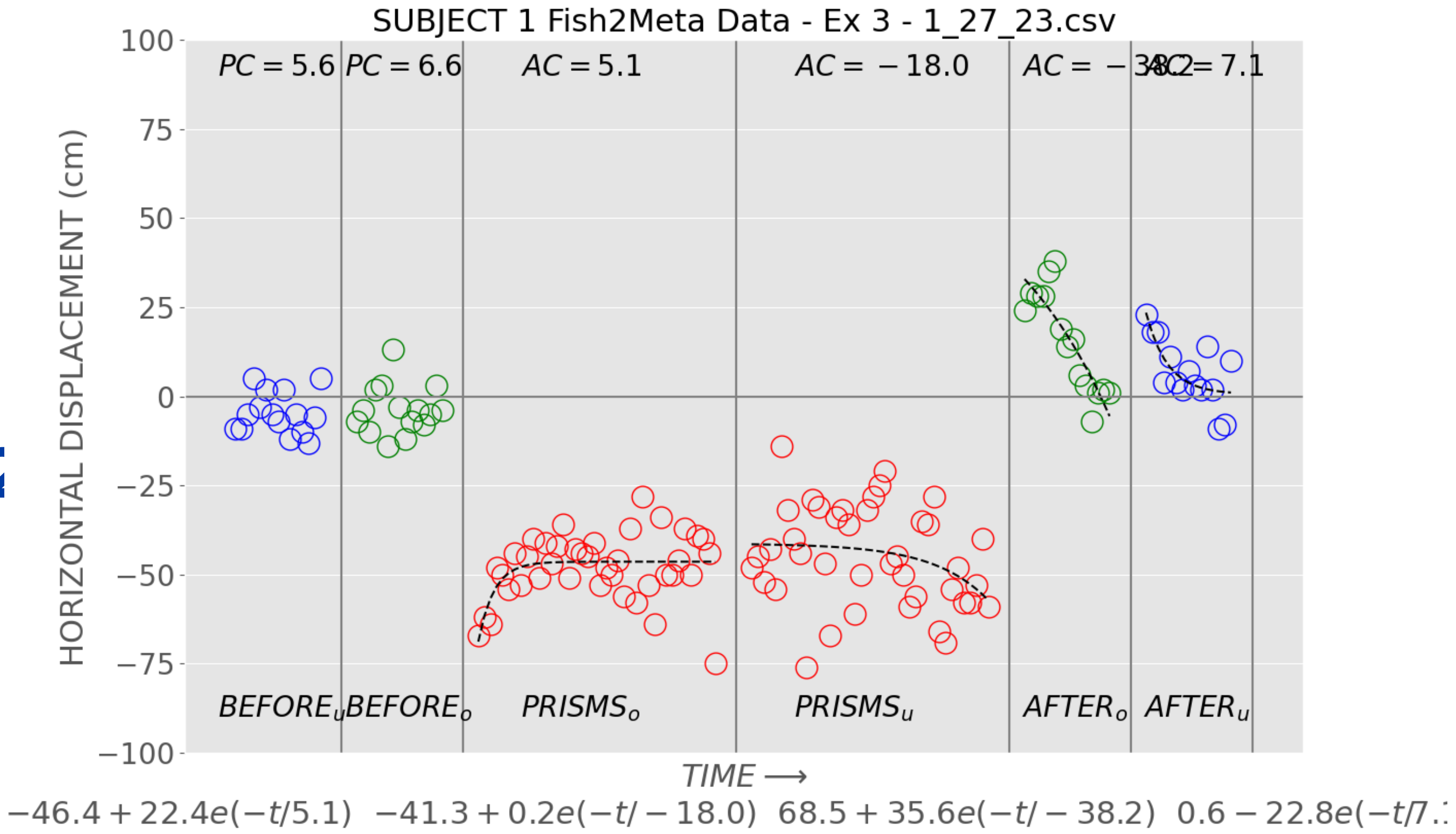
What do we predict we will see?



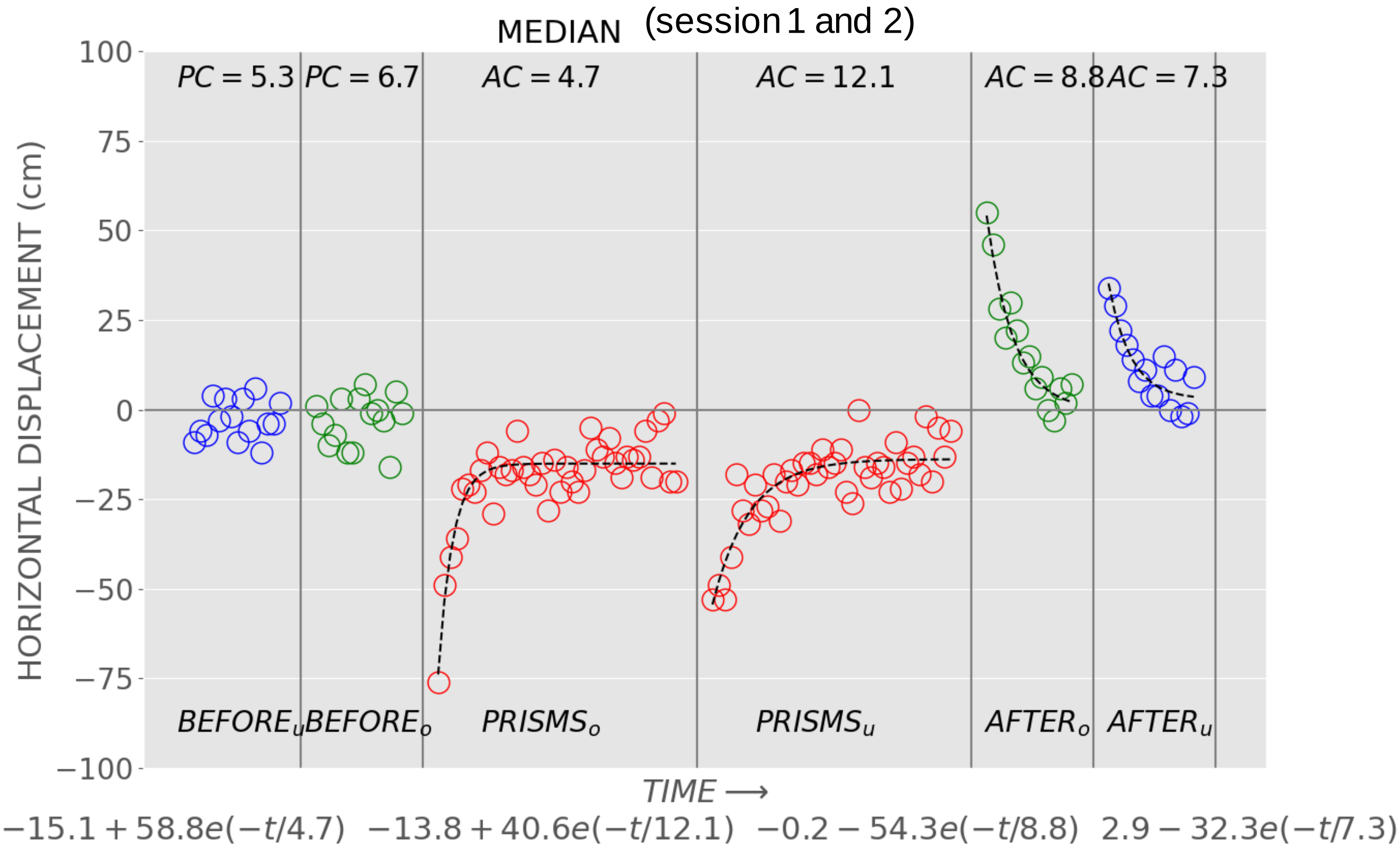
Results session 2



Results session 2



Median results
(n=6)



Conclusion

- The results confirm our predictions. We can see a learning movement both under and over during the prism adaptation.

- This learning has an impact on the post effect which where the participants had to readapt their throws.

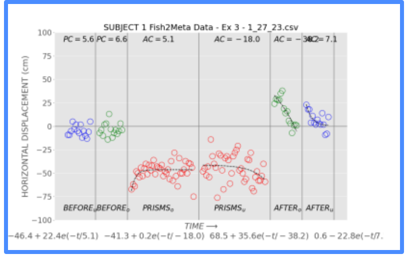
- However, we can notice that the first prism adaptation (over) and its post effect has bigger pic that the second prism adaptation and post effect – there might be some effect.

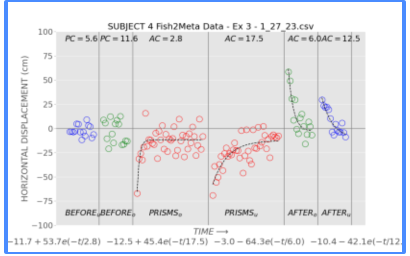
04
Look into the future

Upcoming Goals

Continuing research

- **Learn more about certain subjects**
- **Writing research paper and making poster**





Thanks !